DESIGN & FABRICATION OF AIR POWERED PNEUMATIC VEHICLE (4-WHEELER)

A project report

submitted in the partial fulfilment of the

Requirements for the award of

BACHELOR OF TECHNOLOGY

in

MECHANICAL ENGINEERING

Submitted

by

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DEPARTMENT OF MECHANICAL ENGINEERING

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UNIVERSITY COLLEGE OF ENGINEERING SULTANPUR

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CERTIFICATE

This is to certify that the project entitled

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In partial fulfilment of the requirements for the award of Bachelor of Technology in Mechanical Engineering is a record of Bonafide work carried out under our guidance and supervision during the academic year 2024-25. The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

PROJECT GUIDE

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EXTERNAL EXAMINER

DECLARATION

We the undersigned students of final year in B. tech Mechanical Engineering, JNTUH University College of Engineering, Sultanpur, hereby declare that the dissertation entitled **"AIR POWERED PNEUMATIC VEHICLE(4-WHEELER)",** embodies the report of our project work carried out by **Batch-B** independently by us under the guidance of **Mr. A. Manohar**, **Assistant Professor, ME Department, JNTUH-UCES**, in partial fulfilment for the award of Bachelor of Technology in Mechanical Engineering from Jawaharlal Nehru Technology University, Hyderabad during the academic year 2024-2025.

We also declare that to the best of our knowledge and behalf, this project has not been submitted for the award of any other degree on earlier occasion by any student.

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ABSTRACT

In today's world, rising fuel prices and increasing pollution have created an urgent need to explore clean and sustainable alternatives for transportation. Our project, "Fabrication of an Air Powered Pneumatic 4-Wheeler Vehicle," aims to address this challenge by developing a working prototype that runs entirely on compressed air—without using petrol, diesel, or any harmful emissions. This small-scale vehicle uses a 5kg air cylinder to power a pneumatic piston, which is controlled by a 5/2 solenoid valve. The solenoid is operated using a 12V rechargeable battery. The piston's linear motion is converted into rotary motion through a chain and sprocket mechanism, which drives the wheels. The frame is made from mild steel for strength, and nylon wheels are used to keep the weight low. The goal of this prototype is not high speed or long range, but to show how pneumatic power can be practically applied in mobility. It's a clean, low-cost system that can be used in short-distance travel or educational models. Through this project, we've also gained hands-on experience in integrating mechanical, pneumatic, and electrical systems. While the current range and efficiency are limited by the energy capacity of compressed air, this model opens up possibilities for future improvements—like using higher pressure tanks, optimizing valve timing, or even reusing exhaust air. Overall, this project highlights a simple but promising step toward eco-friendly vehicle technology. This prototype emphasizes a low-cost and pollution-free approach to short-distance travel, suitable for educational and conceptual applications. The project explores the integration of mechanical, pneumatic, and electrical systems in a compact and functional design. While range and power are limited by the energy density of compressed air, the vehicle serves as a proof-of-concept for future development in green mobility. The work also discusses performance characteristics, operational challenges, and potential improvements such as multi-stage air expansion and regenerative braking to extend range and efficiency. This project demonstrates how pneumatic energy can be a viable alternative for compact transportation needs.

CHAPTER-1
INTRODUCTION

INTRODUCTION

This fabrication is based on pneumatics which deals with the fabrication and application of pressurized air to produce force to mechanical four-wheeler vehicle motion. Pneumatic actuator is a fabricated model which when installed in four wheels; this fabricated model consists of a pneumatic control valve which regulates the air flow and pneumatics which is driven by the microcontroller connected with relay switch used in four wheels with connecting rod.

Compressed air as a source of energy in different uses in general and as a non-polluting fuel in compressed air vehicles has attracted scientists and engineers for centuries. Efforts are being made by many developers and manufacturers to master the compressed air vehicle technology in all respects for its earliest use by the mankind. Nowadays, almost every industry is trying to develop light and efficient vehicles. Today, all the vehicles running on conventional & non-conventional fuels are known for producing a large amount of harmful gases like CO_2, SO_2, NO_2, etc. which acts increases global warming. The motto of our project is to design & fabricate vehicle running on air pressure for material handling in industries and reduce power consumption.

Features:

1. User-friendly interaction with the device.
2. Highly sensitive.
3. Easily operable.
4. Pneumatics based four-wheeler control.

1. Background

The rapid industrialization and urbanization in the 21st century have led to an exponential increase in the number of vehicles on roads, significantly contributing to environmental pollution and depletion of fossil fuels. Traditional vehicles powered by internal combustion engines (ICEs) emit large quantities of greenhouse gases such as CO_2, NOx, and unburned hydrocarbons, posing a severe threat to both environmental and human health. As the demand for sustainable and clean energy solutions grows, the exploration of alternative propulsion systems has gained momentum.

2. Limitations of Existing Technologies

Existing automotive technologies, particularly those based on fossil fuels, have several limitations despite improvements in fuel economy and emissions control. Hybrid and electric vehicles offer cleaner alternatives, but they face challenges such as high battery costs, limited charging infrastructure, and long recharging times. Compressed Air Vehicles (CAVs), though an emerging concept, have been limited by range, power output, and scalability. Early air-powered vehicles were often designed for low load capacity and simple operation, lacking the efficiency and power necessary for urban or practical use.

3. Need for an Alternative System

To address the above issues, there is a strong demand for an eco-friendly, efficient, and affordable alternative transportation solution, especially in developing nations like India. The solution should ideally be free of toxic emissions, cost-effective in terms of manufacturing and maintenance, easy to operate, and capable of functioning in urban environments with limited infrastructure. In particular, there is a growing opportunity for lightweight, small-capacity vehicles designed for campus mobility, intra-factory travel, and clean transportation in ecological zones.

4. Proposed System – Air Powered Pneumatic 4-Wheeler Vehicle

This project proposes the design and development of an Air Powered Pneumatic 4-Wheeler Vehicle that utilizes compressed air stored in a tank to drive a pneumatic cylinder, which converts air pressure into mechanical motion. This motion is transmitted to the wheels via a chain drive mechanism. The proposed system overcomes many of the limitations of earlier air-powered designs by integrating improved component sizing (stroke length, bore diameter), optimal transmission design, and lightweight materials. It is engineered for urban and closed environment applications.

5. Working Principle

The working principle is based on converting potential energy stored in compressed air into kinetic energy through a pneumatic actuator. When compressed air is released into the double-acting cylinder, the piston moves back and forth. This linear motion is then converted into rotary motion using a chain-sprocket system connected to the wheels. The process is controlled using a solenoid valve and pressure regulator to ensure consistent operation. The air used can be refilled using a standard air compressor, offering quick and easy recharging.

6. Advantages of the Proposed System

• Zero emissions: No combustion of fuel means no harmful gases are released into the atmosphere.

• Lightweight and low-cost: Simple mechanical design using readily available materials.

• Safe operation: No inflammable fuel is used, reducing risk of fire or explosion.

• Ease of maintenance: Fewer moving parts compared to internal combustion engines.

• Modular and scalable: Can be adapted for various light-duty applications.

7. Scope in Indian Context

In India, especially in southern states, the hot climate and dense urban areas present unique challenges to conventional vehicles. This air-powered design, with its lightweight frame (approx. 12 kg), compact size, and zero emissions, is ideal for short-distance travel within campuses, factories, and green zones. It does not rely on grid electricity or fuel stations, making it suitable for semi-rural deployment as well. Moreover, its low-cost construction makes it appealing for educational institutions and small-scale industrial use.

8. Contribution to Engineering Education and Innovation

This project contributes to mechanical engineering education by offering hands-on exposure to pneumatic systems, fluid power applications, energy conversion, and sustainable design principles. It fosters innovation through problem-solving in areas like transmission efficiency, air pressure optimization, and lightweight chassis design. The knowledge gained through this project can also pave the way for future research in hybrid air-electric systems or regenerative pneumatic braking systems.

9. Conclusion

The Air Powered Pneumatic 4-Wheeler Vehicle represents a significant step forward in sustainable vehicle design. By addressing the limitations of traditional and earlier compressed air vehicles, this project provides a feasible and functional solution tailored for low-speed, eco-friendly transport. Its technical simplicity, environmental benefits, and practical adaptability make it a compelling candidate for real-world deployment and further research.

CHAPTER-2

LITERATURE REVIEW

LITERATURE REVIEW

1. Introduction to Air-Powered Vehicles

Air-powered or compressed air vehicles (CAVs) utilize **compressed air as the working medium** to drive mechanical components. The concept, although not new, has gained renewed interest due to the urgent need for **clean, sustainable energy alternatives** in the transportation sector.

- **Madhusudan et al**. (2013) studied the potential of compressed air as a green energy source and concluded it could significantly reduce greenhouse gas emissions in urban transport.
- **Tewari & Joshi** (2015*)* emphasized the safety, non-toxicity, and ease of refuelling as advantages over fossil fuels.

2. Pneumatic Drive Systems

The use of pneumatic actuators (especially cylinders) for propulsion has been explored in light-weight vehicles:

- **P. Kumar et al.** (2016) analysed single-acting pneumatic cylinders for 2-wheeler movement, highlighting low torque and range limitations.
- **Gupta et al.** (2018) improved system efficiency by integrating multi-stage air storage tanks and pressure regulators.

3. Cylinder and Transmission Mechanisms

- Research by **S. R. Kharde et al.** (2017) evaluated different piston sizes and found that **larger bore diameters with optimized stroke lengths** increased power output but required more air volume.
- **M. Singh** (2020) proposed using **chain drive** systems over direct drive for better torque transmission and smoother movement — similar to what you're implementing.

CHAPTER-3
OBJECTIVES

OBJECTIVES

The objectives of our present work are as follows:

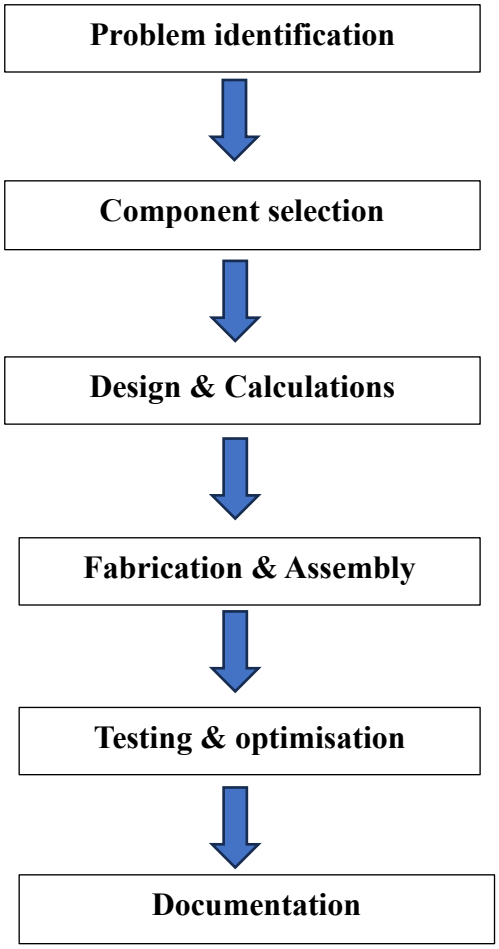
- The objective of this project is to design and develop a lightweight, eco-friendly, and cost-effective four-wheeled vehicle powered entirely by compressed air. The aim is to demonstrate how pneumatic energy can be efficiently used as a clean alternative to conventional fuels for short-distance transportation. This vehicle is intended to operate without producing any emissions, making it suitable for campuses, industrial zones, and pollution-free areas. The design emphasizes simple mechanical construction, ease of operation, and practical application using basic pneumatic components.

- To design a compact and lightweight 4-wheeler vehicle that operates on compressed air as the only power source.
- To demonstrate the application of pneumatic systems in automotive propulsion.
- To eliminate the use of conventional fossil fuels and reduce harmful emissions**.**
- To develop a cost-effective and simple mechanical design using readily available components.
- To analyse and optimize the force transmission and energy efficiency of a pneumatic-driven system.

CHAPTER-4
METHODOLOGY

FLOWCHART

The following flowchart shows the methodology used by us in the construction of the 4-wheeler vehicle.



CHAPTER-5
DESCRIPTION OF COMPONENTS

5.1 MECHANICAL COMPONENTS

5.1.1 CYLINDER

6Kg capacity, made up of mild steel. Used for air storage.

Fig 5.1: Cylinder

5.1.2 PRESSURE GAUGE

A pressure gauge is an essential measuring instrument used to monitor the air pressure within the system, ensuring safe and efficient operation of the pneumatic cylinder. In this project, the gauge helps confirm whether the operating pressure (up to 85 psi or ~6 bar) is being properly delivered from the 10 kg capacity air tank.

- **Key Specifications:**

Parameter	**Description**
Type	Analog (Dial-Type) Bourdon Tube Gauge
Pressure Range	0 – 150 psi (0 – 10 bar)
Dial Size	2 to 2.5 inches (standard readable size)
Connection Type	1/4-inch BSP (Bottom or Back Mount)
Casing Material	Stainless steel or ABS plastic

Fig 5.2: Pressure gauge

5.1.3 BALL VALVE

A **ball valve** is a manual valve that controls the flow of compressed air by using a rotating ball with a hole (bore) in the centre. It is used to start or stop the air supply from the air tank to the pneumatic circuit. Usually made up of Brass.

- **Working Principle:**

- When the valve handle is turned 90 degrees, the hole in the ball aligns with the flow path, allowing air to pass through.
- Rotating the handle again closes the passage**,** blocking the airflow completely.
- This quick shut-off feature makes ball valves ideal for isolating the air tank before system operation.

Fig 5.3: Ball valve

5.1.4 SOLENOID VALVE

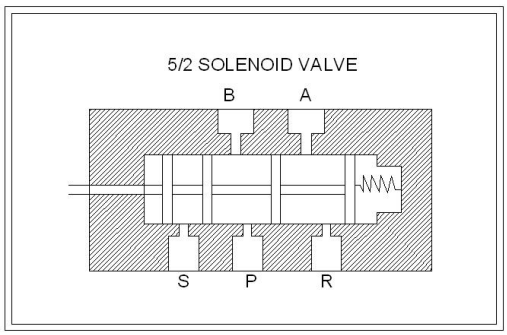


Fig 5.4: Solenoid valve internal view

A 5/2 solenoid valve is used to control the direction of air flow in pneumatic systems, especially for double-acting actuators (cylinders).

The valve has five ports:

- 2 working ports (A and B): Connected to the actuator (e.g., pneumatic motor).
- 2 exhaust ports (EA and EB): For venting air from the actuator.
- 1 pressure port (P): Connected to the compressed air source.
- 2 Positions: The valve has two switching positions that determine the direction of air flow.

Fig 5.5: Solenoid valve

➢ **Working Principle of solenoid valve**

- The valve uses electromagnetic solenoids to switch positions.
- When the solenoid coil is energized, it shifts the spool inside the valve to redirect the flow of air.
- This changes the direction of the actuator (i.e., forward or reverse motion of the vehicle).

▪ **Advantages Solenoid valve in this Project:**

- Precise Direction Control
- Compact and Reliable
- Fast Switching

▪ **Application in this project**

- Used to control the direction of air flow to the pneumatic actuator/motor.

- Enables the vehicle to move forward and reverse by toggling the valve.
- Can be integrated with electrical control circuits or switches for easy driver operation.

- **Parts of a solenoid valve**

1. Coil

The solenoid coil is made of copper wire. The layers of wire are separated by insulating layer. The entire solenoid coil is covered with a varnish that is not affected by solvents, moisture, cutting oil or often fluids. Coils are rated in various voltages such as 115 volts AC,230volts AC,460volts Ac,575 Volts AC.6Volts DC,12Volts DC,

24 Volts DC,115 Volts DC &230Volts DC.
They are designed for such Frequencies as
50Hz to 60Hz.

2. Frame

The solenoid frame serves several purposes. Since it is made of laminated sheets, it is magnetized when the current passes through the coil. The magnetized coils attract the metal plunger to move. The frame has provisions for attaching the mounting. They are usually bolted or welded to the frame. The frame has provisions for receivers, the plunger. The wear strips are mounted to the solenoid frame, and are made of materials such as metal or impregnated less Fiber cloth.

3. Solenoid plunger

The solenoid plunger is the mover mechanism of the solenoid. The plunger is made of steel laminations which are riveted together under high pressure, so that there will be no movement of the lamination with respect to one another. At the top of the plunger a pin hole is placed for making a connection to some device. The solenoid plunger is moved by a magnetic force in one direction and is usually returned by spring action. Solenoid operated valves are usually provided with cover either the solenoid or the entire valve. This protects the solenoid from dirt and other foreign matter, and protects the actuator. In many applications it is necessary to use explosion proof solenoids.

5.1.5 PNEUMATIC CYLINDERS

It consists pistons made up of aluminum alloy, outer layer is made up of stainless steel. The pneumatic piston is the heart of the motion system in your vehicle. It directly translates the power of compressed air into motion, allowing your air-powered car to move forward.

- **Working Principle:**
  - Pneumatic cylinders operate using pressurized air supplied from a tank. When air enters one side of the piston inside the cylinder, it pushes the piston rod outward (or inward), producing linear motion. The direction of the piston movement is controlled using a **5/2 solenoid valve**.

- **Types of Pneumatic Cylinders:**

  1. **Single Acting Cylinder (SAC):** Air pressure moves the piston in one direction, and a spring returns it.
  2. **Double Acting Cylinder (DAC):** Air pressure moves the piston in both directions. Our project uses a **Double Acting Cylinder** for continuous to-and-fro motion, which enhances drive performance.

Parameters

Type	:	Double acting
Stroke length	:	100mm
Bore diameter	:	25mm
Operating pressure	:	10 Bar
Material	:	Aluminum alloy

Fig 5.6: Pneumatic cylinders

5.1.6 PNEUMATIC TUBING PIPES

Pneumatic tubing pipes are flexible hollow tubes used to transport compressed air from the storage tank to various components like solenoid valves and pneumatic cylinders. They act as the arteries of a pneumatic system, enabling smooth and controlled airflow throughout the vehicle.

- **Key Features:**

- **Material:** Common materials include **Polyurethane (PU), Polyethylene (PE), Nylon (PA), or PVC**. PU is preferred for its flexibility and durability.
- **Pressure Rating:** Must be capable of withstanding operating pressures up to 100 psi or more, depending on the system.
- **Diameter:** Generally, ranges from **6 mm or 8 mm OD** tubes are suitable.
- **Temperature Resistance:** Should handle ambient temperatures between **-20°C to +60°C**.
- **Flexibility:** Essential for routing through tight corners without kinking.

Fig 5.7: Pneumatic tubing pipes

5.1.7 CHAIN & SPROCKET

- **Chain**

We have used a simple bicycle chain for this project. A bicycle chain is a mechanical component used to transmit rotary motion and torque between the driving and driven sprockets. In this project, it plays a vital role in transferring the **linear motion from the pneumatic cylinder** into **rotational motion of the wheels**.

Key Features:

- Made of interconnected steel links that form a flexible yet strong loop.
- Designed to run smoothly over toothed sprockets, maintaining grip and motion synchronization.
- Provides efficient power transmission with minimal slippage.
- Can easily be adjusted in length and tension for proper alignment.

Application in the Project:

- Connects the crank or lever arm (attached to the piston) to the rear axle sprocket.
- Converts the back-and-forth motion of the pneumatic cylinder into continuous wheel rotation.
- Helps achieve desired speed and torque by selecting appropriate sprocket sizes.

Fig 5.8: bicycle chain

- **Sprocket**

A sprocket is a toothed wheel that is used to transmit motion and torque from one shaft to another. Chains that are used to transmit motion and force from one sprocket to another are called **power transmission chains.** Unlike gears that have to mesh to transmit motion and torque from one gear to another, sprockets may be positioned far apart. Sprockets are connected by a chain.

A sprocket with a hole that matches the diameter of the shaft is chosen and slid onto the shaft. The sprocket is then rotated until the keyways in the sprocket and shaft line up. A rectangular piece of steel bar called a key is slid into the keyways to prevent the sprocket from spinning on the shaft.

Fig 5.9: Sprocket

- Chain and sprocket mechanisms perform the same task as a belt and pulley system, i.e. they transfer motion and force from one shaft to another. A belt can slip on a pulley but the teeth on the sprocket prevent the chain from slipping. A chain and sprocket is used wherever a positive, non-slip drive is required, e.g. bicycles, motorcycles, forklift mechanisms, and the camshaft drives in car petrol engines.

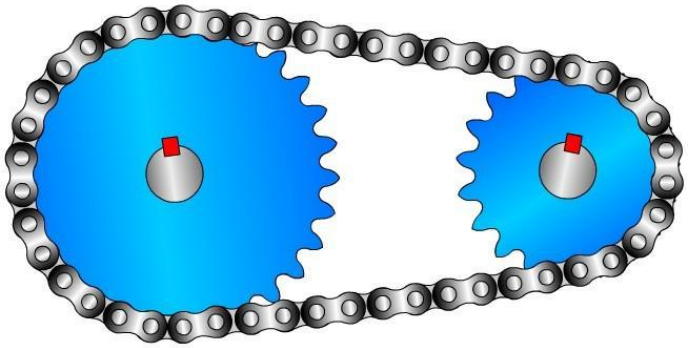


Fig 5.10: Chain & Sprocket mechanism

- **Advantage of chain drives**

An advantage of chain drives over most belt drives is that the chain cannot slip on the sprocket, so the chain and sprocket provide a positive, non-slip drive, i.e. the chain cannot slip on the sprocket because the sprocket teeth prevent the chain from slipping.

Some belt and pulley drive also have teeth. These toothed belt and pulley drives are used in applications where it is important that the belt does not slide on the pulley, e.g. timing belts in internal combustion engines and the drive belts that replace the chain on some motorcycles.

- **Disadvantage of chain drives**

The disadvantage of the chain and sprocket drive is that it can be noisy and more expensive than a belt and pulley drive system.

Chain and sprocket drives are not used on some applications such as drilling machines and lathes precisely because the chain and sprocket drive does not allow slip. For safety reasons, belt and pulley drives are used on many machines so that in the event of something jamming in the machine, under great pressure, the belt can slip on the pulley rather than damaging the machine as would happen with a no-slip chain and sprocket drive.

Chain and sprocket drives are used in applications where slipping should not occur, e.g. bicycles and motor cycles cam shaft drives in internal combustion engines fork lift trucks elevators and escalators.

5.1.8 DOUBLE THREADED HUB

This sprocket is mainly connected through a double threaded hub. The hub is made up of stainless steel and welded that chain should not slip from the thread. A double threaded hub is a specialized wheel hub that features threads on both sides of its central axle or shaft, allowing two sprockets (or a sprocket and a freewheel) to be mounted—one on each side. This type of hub is commonly used in bicycles and small mechanical vehicles to facilitate bidirectional power transfer or alternate drive mechanisms.

- **Key Features:**

- Allows mounting of two sprockets or gear sets on a single hub.
- Made of hardened steel or aluminium alloy for strength and durability.
- Often found in fixed-gear or dual-drive systems.

- **Application in This Project:**

- In our pneumatic 4-wheeler, the double threaded hub is mounted on the rear axle.
- Drive sprocket connected to the chain from the pneumatic piston mechanism.
- Enables better alignment and simplifies power transmission, especially when the drive source is reciprocating.

- **Advantages:**

- Compact design—supports dual-drive setup without extra shafts.
- Ensures strong mechanical grip with threaded engagement.
- Low maintenance and easy to install or replace sprockets.

Fig 5.11: Double threaded hub

5.1.9 CHASSIS FRAME

The chassis frame is the foundational structure of the vehicle, responsible for supporting all mechanical and pneumatic components, including the wheels, air cylinder, drive system, and control units. In this project, the frame is fabricated using mild steel due to its strength, durability, and ease of welding and machining. Nylon plastic tyres are used to move, which are light weight and better durability.

- **Material Used:** Mild Steel (MS)

Mild steel is a low-carbon steel known for its:

- Good weldability.
- Sufficient strength for light-load applications.
- Cost-effectiveness.
- Resistance to bending under moderate stress.

Fig 5.12: Chassis frame

- **Frame Dimensions:**

Parameter	Value
Total Length:	24 inches
Length Member Thickness:	1 inch
Total Breadth:	14 inches
Breadth Member Thickness:	0.5 inch

- **Advantages of This Frame Design:**

- Strong yet lightweight, suitable for the overall 12 kg vehicle weight.
- Provides rigid support for all components with minimal vibrations.
- Customizable and scalable for future improvements or testing.
- Simple geometry ensures easy fabrication in workshops with basic tools.

5.2 ELECTRICAL COMPONENTS

5.2.1 BATTERY

SANCA brand Sealed Lead-Acid (SLA) Rechargeable Battery with a 12V, 7.5Ah capacity is used as the primary power source for all electrical components, including the solenoid valve, control switches, and optional indicators.

- **Key features**

Specification	Value
Battery Type	Sealed Lead-Acid (SLA)
Brand	SANCA
Voltage Rating	12 Volts DC
Capacity	7.5 Ampere-Hour (Ah)
Rechargeability	Yes (Maintenance-free)
Output Terminals	T1/T2 spade connectors or screw type
Operating Temperature	-15°C to +50°C

- **Function in the Project:**

- Powers the 5/2 Solenoid Valve, enabling directional control of the pneumatic cylinder.
- Can also supply LED indicators, relays, and switches used in the vehicle's control system.

- **Advantages:**

- Compact and lightweight
- Maintenance-free
- Reliable discharge performance
- Cost-effective and readily available.

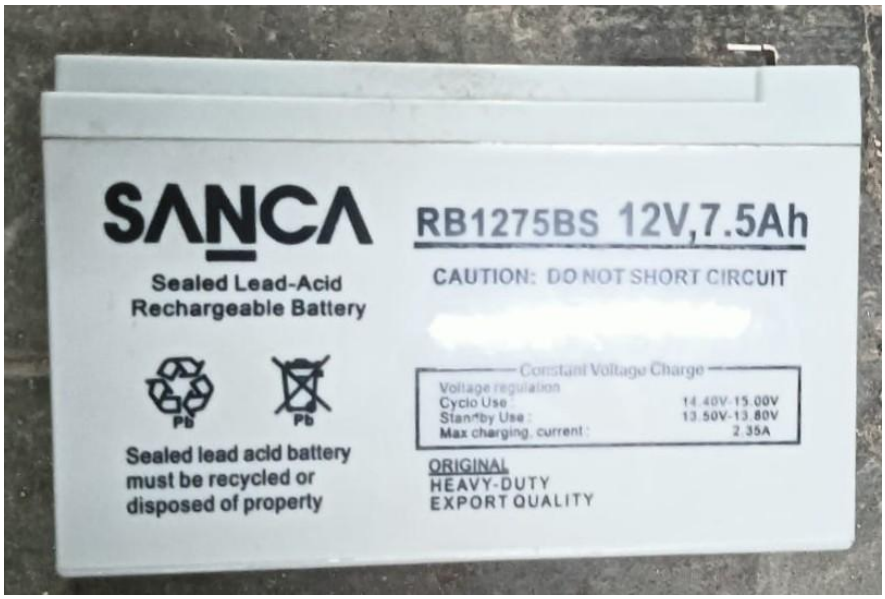


Fig 5.13: Battery

4.2.2 Switch

A simple switch is used to control the solenoid valve from battery connection, the switch is connected through standard copper wiring.

Fig 5.14: Switch

5.3 PART LIST

SNO	DESCRIPTION	QTY
1	Cylinder 6kg capacity	**1**
2	Pressure gauge 10kg capacity	**1**
3	Ball valve	**1**
4	Pneumatic cylinder	**2**
5	Pneumatic tubing pipe	**1meter**
6	Bicycle chain	**2**
7	Sprocket	**2**
8	Double threaded hub	**1**
9	Solid led-acid Battery (12v 7.5Ah)	**1**
10	Pneumatic tee connectors	**4**
11	Copper wiring	**1meter**
12	Switch	**1**
13	5/2 Solenoid valve	**1**

Table 5.1 List of parts

CHAPTER-6

DESIGN

DESIGN

This chapter will discuss the design of the low-cost pneumatic 4-wheeler vehicle and the parts required for its fabrication. The design of the model has to be done in software where the actual model with the required dimensions is developed so that it can be used to fabricate the model. To develop and fabricate the model, there are many processes and parameters involved mainly the design of the model. The design process started by keeping in mind that we are making a low-cost pneumatic vehicle. 61 x 35 x 4 mm chassis frame is selected.

- The mechanism of chain & sprocket with the help of double threaded hub is obtained by fitting in a particular way that can be worked with minimal efforts.
- For the cylinder mounting, we have designed an arc shaped holder which can hold the cylinder for better weight distribution and handling.

SOFTWARE USED: FUSION 360

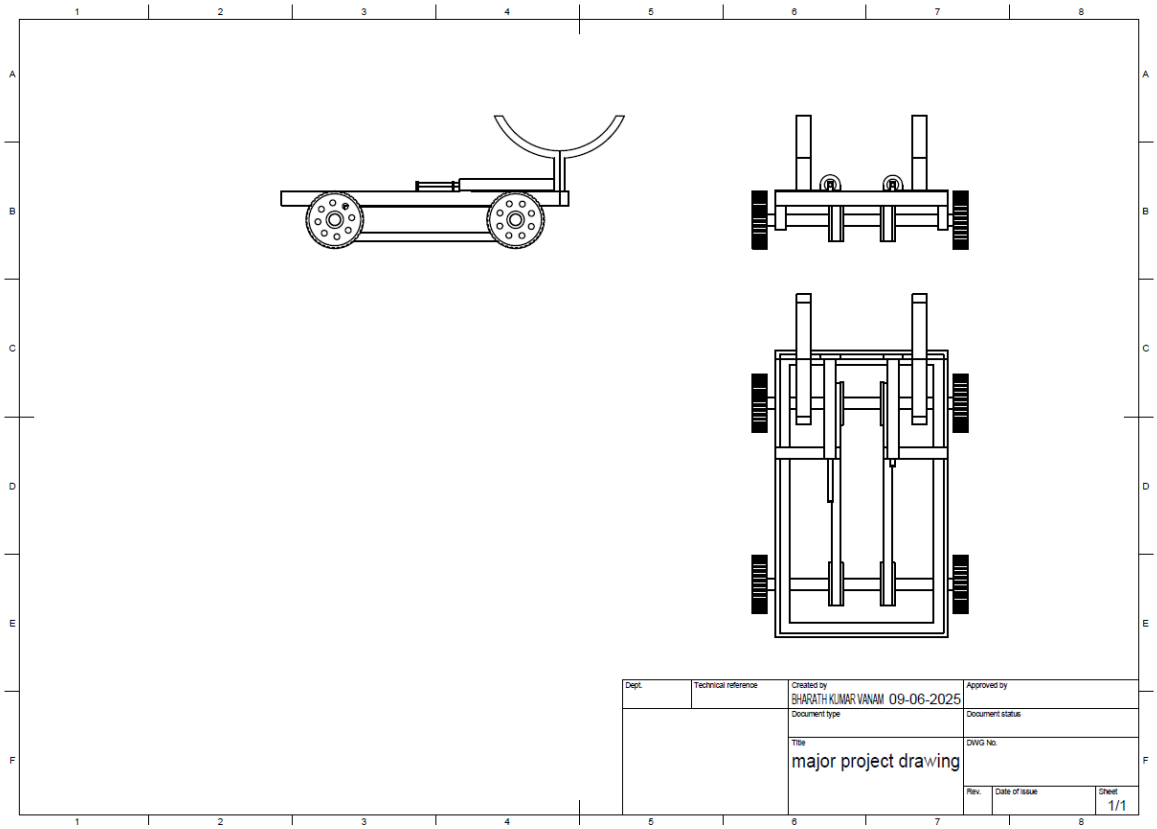


Fig 6.1: Base design of the model

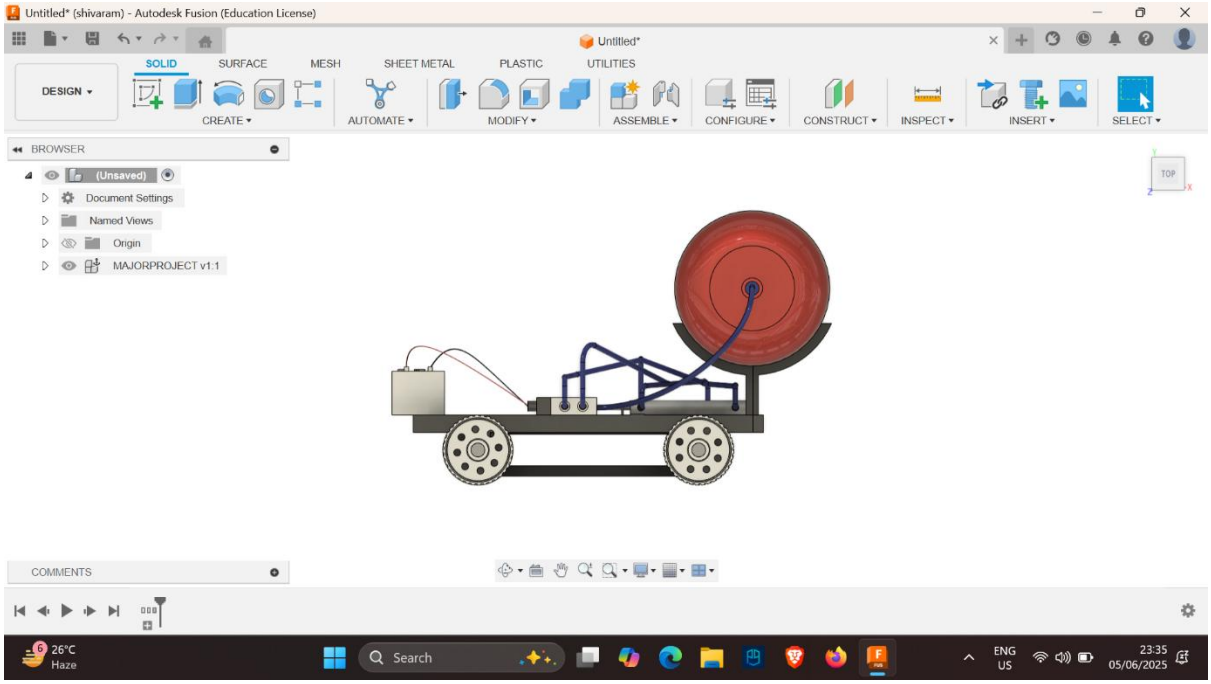


Fig 6.2: Side view of the model

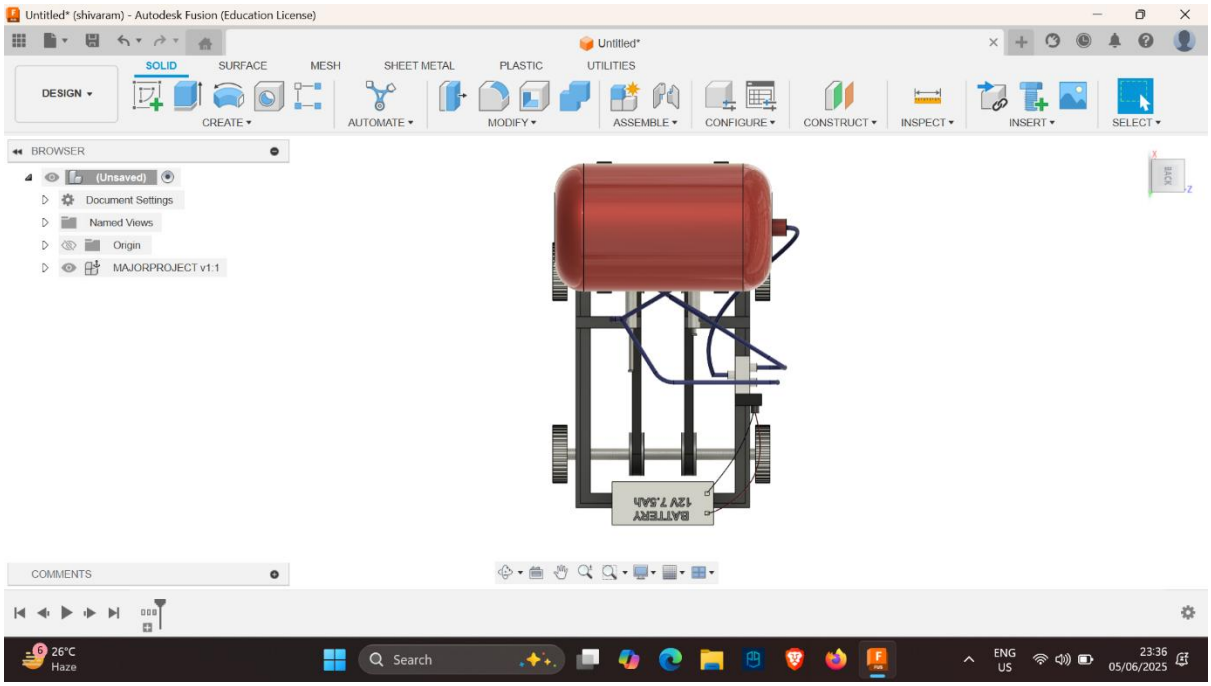


Fig 6.3: Top view of the model

CHAPTER-7
FABRICATION

Steps involved in the fabrication process

Step 1: Conceptual Design and Planning

- Identified the objective: To design a prototype vehicle powered by compressed air.
- Collected necessary data: piston size, pressure range, wheel size, vehicle weight.
- Designed a basic layout including chassis, air tank placement, cylinder mount, and drivetrain system.
- Selected components based on load calculation and pneumatic system design.

Step 2: Frame Fabrication

- Material selected: Mild steel (MS) rectangular and square hollow pipes for light weight and strength.
- Cut MS pipes to required dimensions using a cutting machine.
- Welded the frame into a rectangular base platform to support the components.
- Added brackets and mounts for pneumatic cylinder, tank, and wheel hubs.

Step 3: Mounting the Wheels and Drive Mechanism

- Used nylon plastic wheels (lightweight) with 90 mm diameter.
- Fixed dual-threaded hubs to support chain and sprocket mechanism.
- Mounted rear axle and sprocket system with proper chain tension.
- Ensured smooth rotation and alignment of wheels.

Step 4: Installation of Pneumatic Cylinder

- Selected cylinder: Single/double-acting pneumatic cylinder (25 mm bore, 100 mm stroke).
- Fixed the cylinder to the chassis using clamps and supports.
- Connected piston rod to the drivetrain via a lever/slider-crank or direct chain actuator setup to transmit linear to rotary motion.
- Ensured cylinder moves freely along its axis without friction.

Step 5: Integration of Compressed Air System

- Mounted air tank (6kg) securely on the chassis.
- Connected the tank to the 5/2 solenoid valve using pneumatic tubing.
- Used PU tubing rated for 100+ psi for safe and flexible air transfer.
- Ensured all fittings are airtight using Teflon tape or pneumatic thread sealant.

Step 6: Wiring and Electrical Setup

- Used a 12V, 7.5 Ah lead-acid battery to power the solenoid valve.
- Connected battery to a switch control unit (manual or relay).
- Wired the 5/2 solenoid valve to control piston forward and return strokes.
- Ensured proper insulation and fuse protection for safety.

Step 7: Testing the System

- Checked for air leaks in all tubing and fittings using a soap water test.
- Pressurized the system to **85 psi** using an external compressor.
- Activated the switch to control the solenoid and verified piston movement.
- Observed the vehicle's forward motion and drivetrain performance.

Step 8: Optimization and Troubleshooting

- Adjusted piston alignment and solenoid timing for better performance.
- Checked wheel alignment, friction losses, and improved rolling efficiency.
- Added safety valve or pressure relief if needed.

Step 9: Documentation and Presentation

- Recorded parameters: stroke length, pressure, speed, and range.
- Prepared data sheets, performance charts, and photos for project review.
- Created PPT and report including: design, working, challenges, and outcomes.

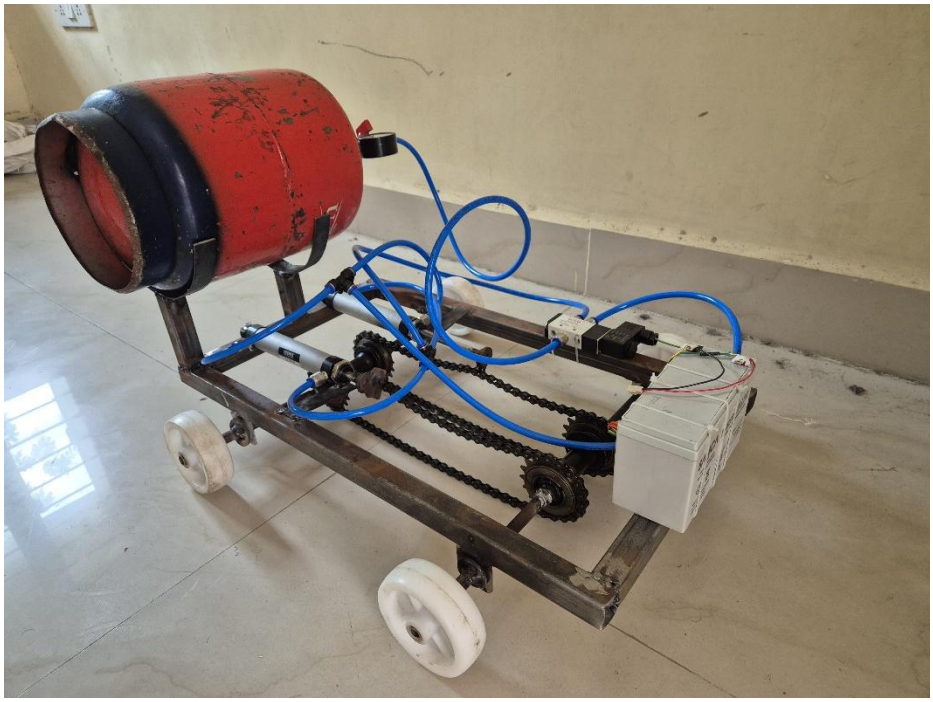


Fig 7.1: Final assembly

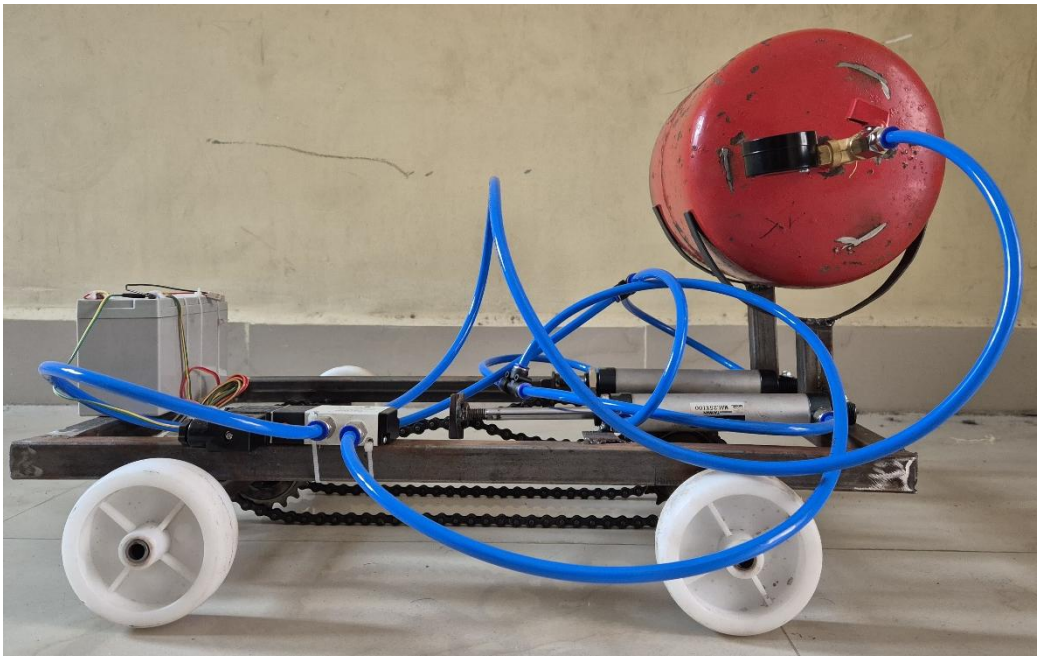


Fig 7.2: Final assembly

CHAPTER-8

RESULT

CALCULATIONS

1. Energy Stored in the Air Tank:

The 6 kg cylinder stores about 23.8 kilojoules of usable energy at 100 bars after considering efficiency losses.

2. Energy Used per Stroke:

Each pneumatic cylinder stroke uses approximately 29.5 joules of energy.

Since we use two cylinders, the energy per cycle is roughly 59 joules.

3. Total Number of Strokes:

Dividing the total energy (23,833 J) by energy per cycle (59 J), we get around 404 piston cycles.

4. Distance Moved per Stroke:

Based on the wheel size and assumed transmission efficiency, each cycle moves the vehicle by around 14 cm.

5. Estimated Range:

Total range = 404 cycles × 14 cm = ~**56 meters.**

This is a theoretical estimate. Actual range will be lower due to:

- Leakage
- Friction losses
- Rolling resistance
- Load
- Incomplete expansion of air

We have tested the actual range of the Vehicle, and it is **28 meters.**

The obtained efficiency is 50%

We have successfully done the testing and analysis of our project vehicle.

CHAPTER-9
CONCLUSION

CONCLUSION

- The fabrication of an Air Powered Pneumatic 4-Wheeler Vehicle was carried out successfully as a part of our final year mechanical engineering project. The main objective of this project was to develop an eco-friendly, cost-effective, and simple vehicle prototype powered by compressed air, which can serve as an alternative to fuel-powered transport systems in the future. The project focused on using pneumatic power to drive a vehicle without any combustion. This project was based on the principle of converting compressed air energy into mechanical energy using a pneumatic cylinder. A 5/2 solenoid valve controlled the direction of airflow to the cylinder, which helped in generating forward and backward strokes. These strokes were then converted into rotational motion to drive the wheels through a chain and sprocket mechanism. A lead-acid battery was used to power the solenoid valve, while PU tubing and fittings ensured safe and controlled airflow. The vehicle was tested at an operating pressure of 85 psi, and it showed satisfactory results in terms of motion and power delivery for a small-scale prototype. The wheels were made of nylon plastic to reduce weight, and a mild steel frame provided good structural support. Despite being a prototype, the vehicle could move under its own power and demonstrated the basic working of a compressed air powered system for industries. One of the main achievements of this project is the successful demonstration of a zero-emission drive system. It also helped us understand the working of pneumatic systems, solenoid valves, and mechanical power transmission. During the fabrication process, we faced challenges related to air leakage, solenoid timing, and mechanical alignment, but these were resolved through practical adjustments and testing. This project highlights the potential of using compressed air as an alternative energy source for light-duty transport systems. Though the current range and efficiency are limited due to air storage capacity and losses.
- In conclusion, this project provided valuable hands-on experience in fabrication, design thinking, and practical problem-solving. With further development and research, compressed air vehicles may become a practical and sustainable option for short-distance urban transport.

CHAPTER-10

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